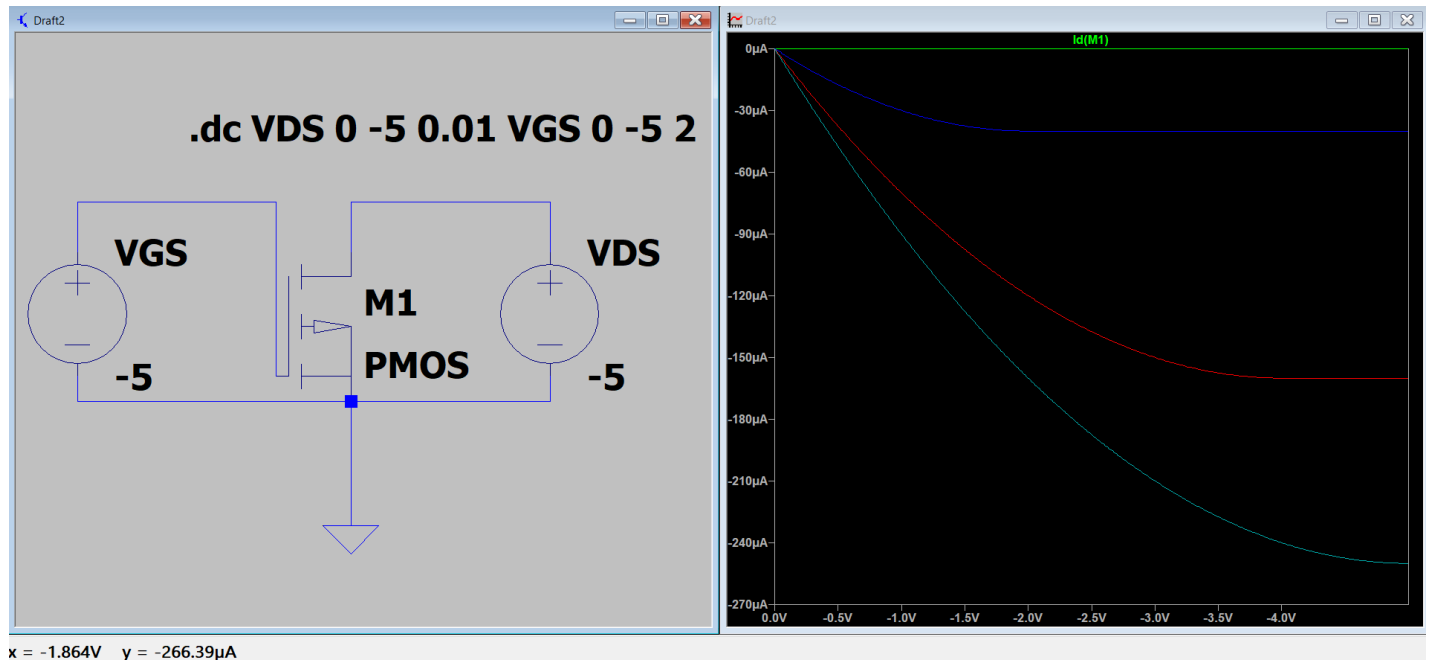
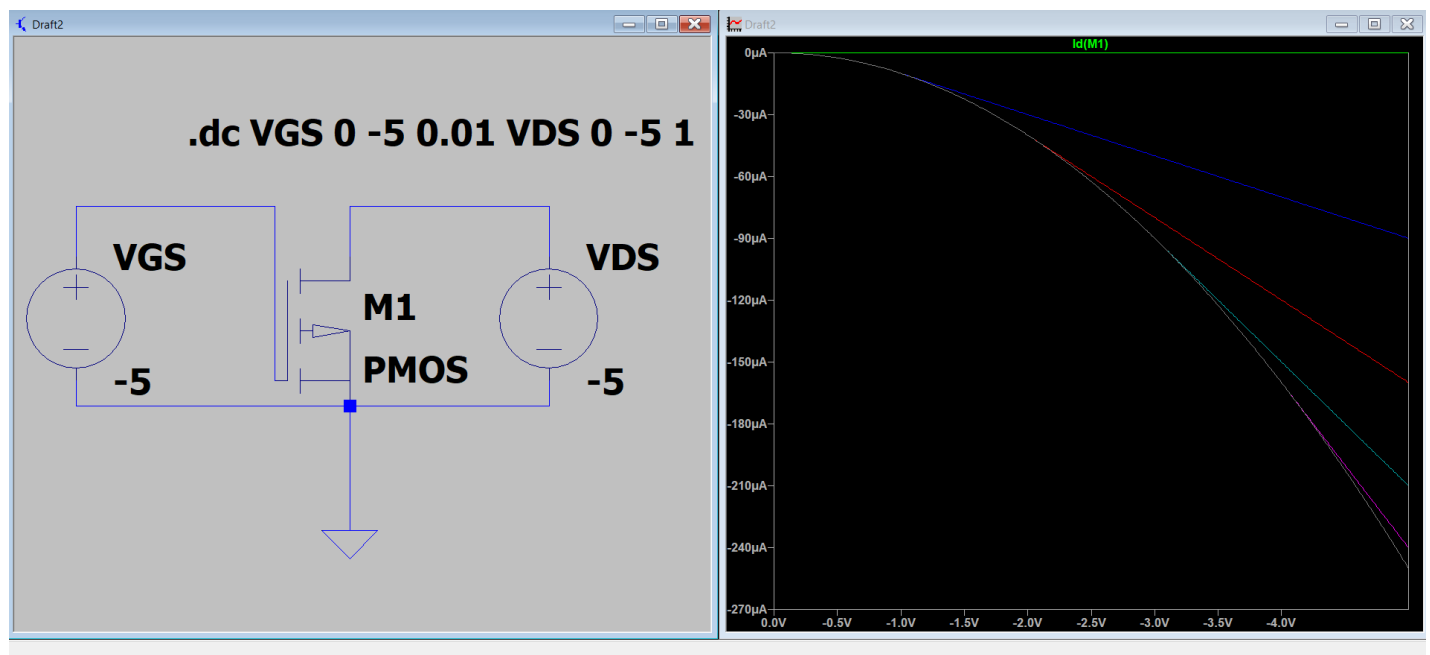


Question 01

PMOS V_{DS} vs I_D (for different V_{GS})

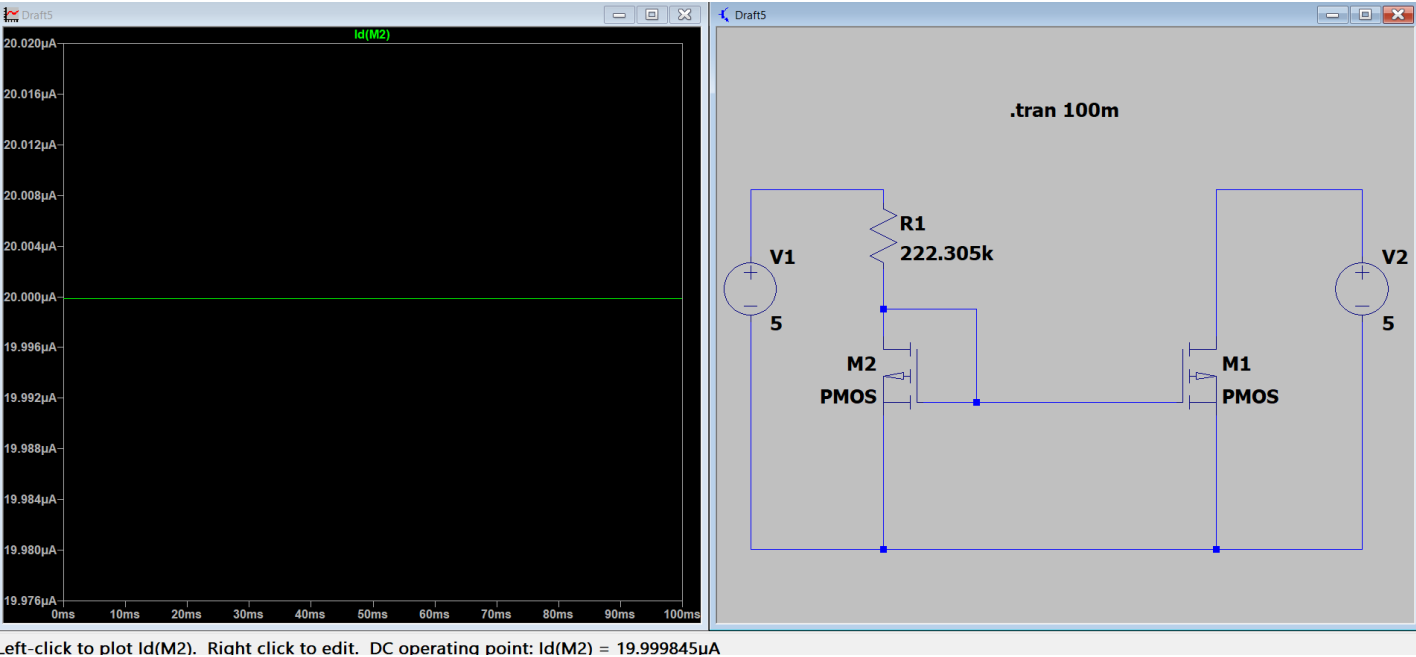


PMOS V_{GS} vs I_D (for different V_{DS})



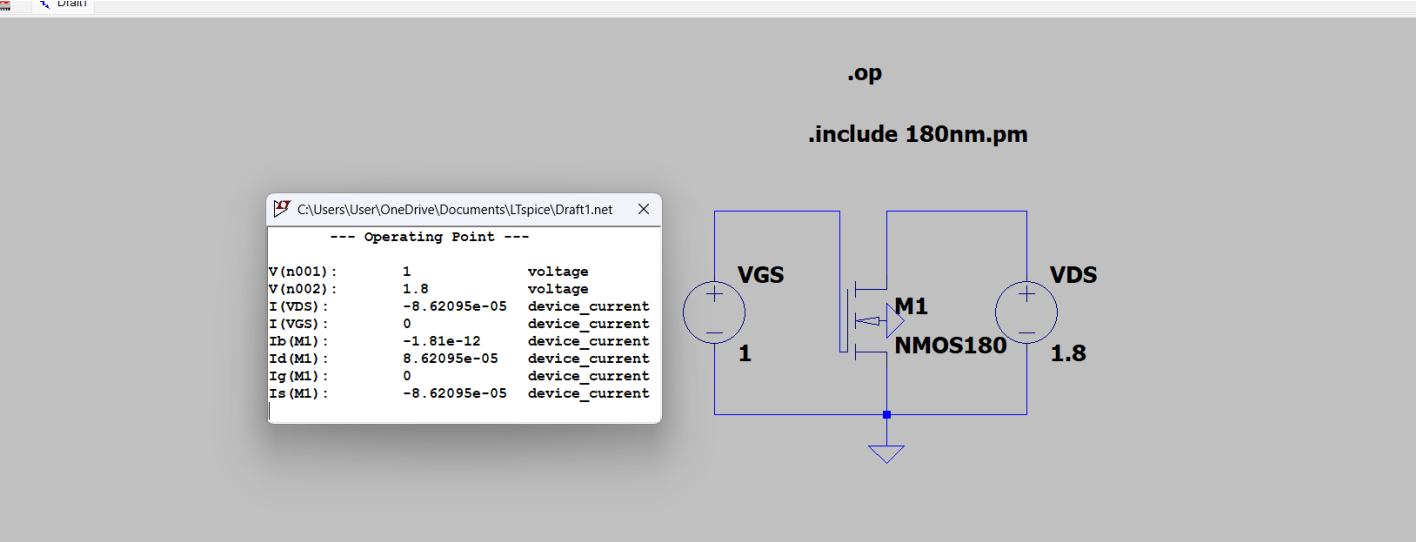
QUESTION 02 .b

PMOS current mirror circuit



Transistor Value: 222.305kΩ

Question 03



```

LTspice 26.0.2 for Windows
Circuit: C:\Users\User\OneDrive\Documents\LTspice\Draft1.net
Start Time: Thu Jul 2 20:12:44 2026
Options: ALLOW_AMBIGUOUS_MODELS
C:\Users\User\OneDrive\Documents\LTspice\180nm.pm(19): Ignoring unknown model parameter. This may or may not be a problem
+xbw= 0 binflag= 0
+Dwg= 0.00 Dwb= 0.00
^^^^^^^^^^^^
C:\Users\User\OneDrive\Documents\LTspice\180nm.pm(79): Ignoring unknown model parameter. This may or may not be a problem
+binflag= 0 Dwg= 0.00 Dwb= 0.00
^^^^^^^^^^^^^^^^^^^^^^^^^^^^
C:\Users\User\OneDrive\Documents\LTspice\180nm.pm(83): Ignoring unknown model parameter. This may or may not be a problem
+trsc= 0 rdc= 0
^^^^^^^^^^^^^^^^^^^^^^^^^^^^
C:\Users\User\OneDrive\Documents\LTspice\180nm.pm(83): Ignoring unknown model parameter. This may or may not be a problem
+trsc= 0 rdc= 0

+K1= 0.5560000 K2= 0.00
^^^^^^^^^^

Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Ignoring BSIM parameter CTA
Ignoring BSIM parameter CTP
Ignoring BSIM parameter PTA
Ignoring BSIM parameter PTF
solver = Normal
Maximum thread count: 12
tnom = 27
temp = 27
method = trap
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.
Direct Newton iteration succeeded in finding operating point.
Total elapsed time: 0.132 seconds.

Files loaded:
C:\Users\User\OneDrive\Documents\LTspice\Draft1.net
C:\Users\User\AppData\Local\LTspice\lib\cmp\standard.mos
C:\Users\User\OneDrive\Documents\LTspice\180nm.pm

```